Eric Hsu

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GitHub: github.com/bshwao

EXPERIENCE

McGill Formula Electric — Electronic Systems Team Member

Montréal, QC | Sept 2025 – Present

- Designed and tested 4-layer PCBs in KiCad for the low-voltage system of a Formula SAE electric race car, cutting board area by 21% while meeting signal-integrity requirements.
- Integrated 6 sensors and built a CAN network across 8 nodes, eliminating dropped messages during track testing.
- Diagnosed and resolved 5 vehicle-wide electrical faults with an over 50+ person multidisciplinary team.
- Verified electrical systems against Formula SAE technical inspection requirements and passed electrical inspection on the first attempt.

PROJECTS

flowst8 — Contributor, team project (3 people) | Hack The Hill III, September 2026 | [https://github.com/ric-e/flowst8]

- Developed a real-time productivity web application designed to calculate dynamic focus scores, detect developer cognitive fatigue, and prompt timely breaks to prevent burnout.
- Engineered an end-to-end data pipeline using Python, FastAPI, and WebSockets to synchronize high-frequency keystrokes per minute (KPM) and live Apple Watch heart rate telemetry.
- Containerized microservices using Docker, implementing TimescaleDB for continuous time-series data storage and the ElevenLabs API for triggering automated, natural-sounding audio alerts.
- Completed the full-stack integration of the HTML/Next frontend with Python/Typescript backend

UNIX CLI Shell Simulator (Test-Shell) | [https://github.com/bshwao/Test-Shell]

- Developed a lightweight UNIX command-line interpreter in C to simulate core operating system interactions and essential shell functionalities.
- Leveraged POSIX system calls (such as fork, execvp, and wait) to manage process creation, child process execution, and process synchronization.
- Created input parsing and memory management protocols to accurately handle command arguments, prevent memory leaks, and mitigate zombie processes.

EDUCATION

McGill University — Montréal, QC
Bachelor of Engineering, Electrical Engineering *(Expected 2030)*

Richmond Hill High School — Richmond Hill, ON
Ontario Secondary School Diploma, 2025

- GPA: 97.5%
- Honor Roll (Grades 9–12)

LANGUAGES & TECHNOLOGIES

Languages: Python, C, C++, Bash/Shell Scripting, Java, JavaScript, TypeScript, HTML, Docker, Node.js, Linux OS, UNIX OS, SQL, Git, Verilog (SystemVerilog), UVM, CARLA, CAD

Engineering / Technical: Circuit Analysis, Digital Logic Design, Microcontrollers, PCB Design, Electronic Testing & Troubleshooting, Oscilloscopes & Multimeters, Technical Documentation, Hardware-Software Integration, Breadboard, Memory Controllers

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ACTIVITIES

Volunteering:

- McHacks 13 (500+ participants)
- McGill TechFair (1000+ participants)

Level 10 Piano with Honors — Royal Conservatory of Music

LFS101 Certification — Linux Foundation

LANGUAGES

- English
- Mandarin
- French